PRA-SGLang: Query-Addressed Non-Prefix Memory beside RadixAttention

Jorge Simão
EInnovator R&D

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Abstract

SGLang's RadixAttention reuses exact token prefixes. Progressive Retrieval Attention (PRA) selects versioned non-prefix resources and intervals according to the current query. We implement a logical PRA object namespace and an SGLang adapter that remains an E0 facade until an installed executor consumes selected native K/V. A 25-request Apple-M5 smoke study with patched SGLang MLX separates exact radix reuse from selected-memory recovery. Prefix plus selected memory and full context both recover the fixed answer in 5/5 requests, while the no-evidence controls are 0/5. Selection uses 365 prompt tokens instead of 1,577. The scheduler reports 364 cached tokens on warm selected requests and 1,576 on warm full-context requests; warm TTFT is 34.1 and 52.7 ms, respectively. These results demonstrate complementarity between exact prefix reuse and black-box selected context. They do not establish hierarchical non-prefix residency: HiCache was disabled, and no PRA-selected K/V object was consumed by SGLang attention. The engine-native hypothesis therefore remains open behind a strong RadixAttention baseline.

1 Introduction

A prefix cache and a retrieval memory answer different questions. A radix tree asks whether token histories share an exact path. PRA asks which retained resource or interval is useful for a query. A long-running session can use both:

$$\underbrace{\text{stable active dialogue}}_{\text{radix prefix}} + \underbrace{\text{query-selected archive}}_{\text{PRA resource}}.$$

SGLang combines language-model serving with a radix cache and optional hierarchical KV mechanisms [1]. The systems question is not whether selected text can be sent to SGLang; a gateway already does that. It is whether a stable non-prefix PRA identity can share SGLang's physical cache hierarchy and scheduling substrate without being misrepresented as a prefix.

This iteration contributes an engine-independent logical object lifecycle, a capability-honest SGLang adapter, and a measured RadixAttention baseline. It also records a negative scope boundary: neither HiCache nor native selected-K/V execution was active, so deep-integration claims are deferred.

2 Two Address Spaces

The two logical keys are

$$a_{\text{radix}} = (t_1, \ldots, t_n), \qquad a_{\text{PRA}} = (\tau, s, r, v, [i, j), \ell, m, d, p),$$

where the PRA key includes tenant τ, optional session s, resource and version (r, v), source interval $[i, j)$, layer ℓ, model revision m, layout/dtype d, and positional/materialization policy p. Equality in one space does not imply equality in the other.

The black-box baseline selects exact evidence outside SGLang and serializes it into the current turn. RadixAttention may then cache that resulting token prefix. An engine-aware design would instead retain the PRA logical identity while placing its physical detail in GPU, host, or distributed hierarchy.

Table 1: SGLang MLX smoke results. TTFT is milliseconds. Warm cached tokens come from the scheduler log rather than the SSE usage object.

Condition	Exact	Prompt	Cold TTFT	Warm TTFT	Warm cached
None	0%	24	511.8	44.3	23
Prefix	0%	333	97.3	31.6	332
Selected	100%	56	40.8	31.5	55
Prefix + selected	100%	365	43.5	34.1	364
Full context	100%	1577	287.4	52.7	1576

3 Implementation Boundary

`src/pra_hf/engine_memory.py` provides validated states IndexedOnly, OffDevice, Prefetching, Resident, Pinned, Evictable, and Invalid. Resident transitions require physical handles. Selection enforces tenant and authorization scope, and snapshots keep address, logical detail, and resident detail bytes disjoint.

`src/pra_sglang/adapter.py` exposes the OpenAI-compatible server as E0. It advertises automatic prefix caching and selected-text fallback, but no logical references or native K/V. Supplying a `SGLangNativeExecutor` promotes the adapter to E2 and transfers generation, streaming, session close, resource deltas, and block-store access to the executor. This pattern prevents a sidecar namespace from being mistaken for attention integration.

An eventual hierarchical bridge must map PRA identities to an independent HiCache namespace or equivalent. It must not insert arbitrary resources into the radix tree as artificial prefixes. It must also preserve GQA/MQA layout, source positions, masks, consumer layers, and request lifetime.

4 Environment

The host was an Apple M5 MacBook Pro with 16 GB unified memory, Metal 4, and macOS 26.4.1. SGLang revision `ef20fab38a03490e2cdf1b7377145ca3a3f2bfc5` ran under Python 3.11.15, MLX 0.32.2, and mlx-lm 0.31.3. The checkout required a local compatibility patch that restores the loader property expected by `MlxModelRunnerStub`; the patch is part of provenance, not hidden setup.

The model was `Qwen/Qwen3-0.6B` at revision `c1899de289a04d12100db370d81485cdf75e47ca`. The server used one running request, 2,048 total tokens, disabled CUDA-graph backends, and the MLX runner. Startup identified `UnifiedRadixCache` with LRU eviction. HiCache was disabled. The control plane reported `torch_native`, while model inference was delegated to MLX; both facts are retained in the artifact.

5 Method

We used the same synthetic codeword task and serialized prompts as the vLLM and MLX-LM calibration. Twelve distractor records surround one authoritative record. Conditions are no prefix/no PRA, prefix only, selected evidence only, prefix plus selected evidence, and full context. Each is repeated five times. SSE timing provides TTFT; SGLang scheduler logs provide exact new and cached token counts. Tail percentiles are not reported from five samples.

This is Smoke evidence. It measures S0/S1 and the selected-text black-box baseline, not HiCache, native PRA attention, distributed cache affinity, or concurrency.

6 Results

The controls behave correctly: no-evidence conditions are 0/5, and every selected or full-context request recovers the answer. Prefix plus selected memory uses 365 tokens, 76.9% fewer than full context. Its cold TTFT is 43.5 ms versus 287.4 ms; its warm mean is 34.1 versus 52.7 ms.

The cache trace is the strongest mechanism observation. After the first request in each condition, the scheduler reports one new token and 364 cached tokens for prefix plus selected memory. Full context similarly uses one new and 1,576 cached tokens. Selected context therefore does not interfere with exact radix reuse. It also creates a much smaller cached sequence. These numbers measure token-prefix cache reuse, not PRA-resource reuse.

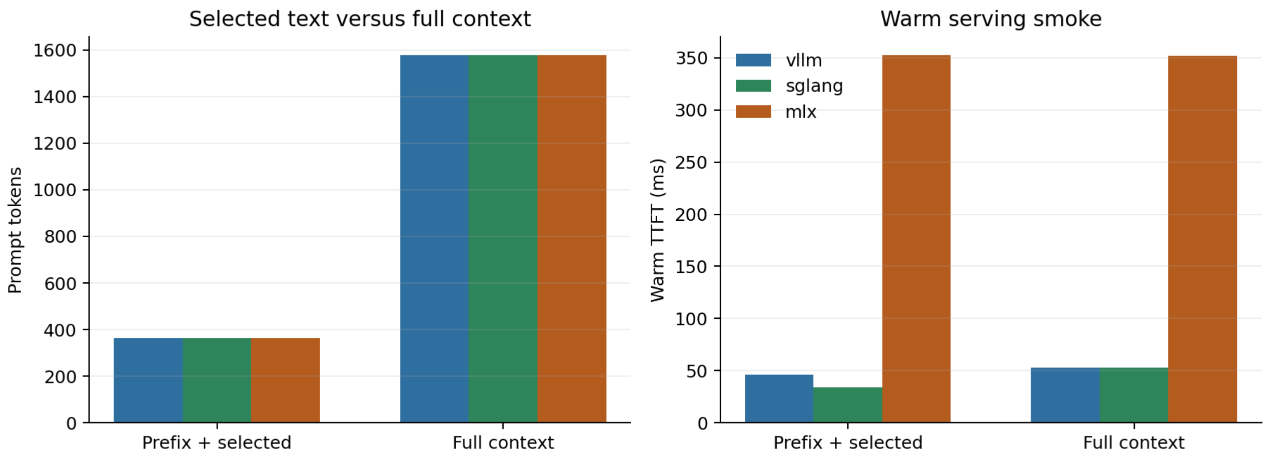

Figure 1: Shared calibration view. Prompt sizes are identical across the Qwen tokenizers. Absolute engine latency is not comparable because SGLang used an unquantized model while vLLM and MLX-LM used a 4-bit checkpoint.

7 What the Result Does Not Show

HiCache was disabled, so no L1/L2/L3 placement, promotion, demotion, transfer, or reload metric exists. The SGLang MLX stub reports zero ordinary GPU KV pool allocation in the control plane because MLX owns inference state. No selected PRA K/V was registered as a hierarchical object, prefetched, shared, or consumed by attention. The present native fields are therefore `NOT_MEASURED`, not zero.

Gate	Status	Evidence
environment	closed	patched pinned SGLang-MLX starts reproducibly
radix baseline	smoke closed	exact scheduler cache-token trace
selected-text parity	smoke closed	selected and full are 5/5
HiCache non-prefix object	open	hierarchy disabled
selected native attention	open	no installed executor
robustness	open	one model, one fixture, concurrency one

8 Next Engine-Native Experiment

The next useful patch is deliberately narrow. It should define an immutable PRA object key beside, not inside, the radix key; place selected detail in host or unified-memory backing; promote it before first use; and feed the selected K/V through an attention path that shares normalization with local K/V. The matched comparison is RadixAttention plus black-box selected text versus Radix plus native PRA at equal quality and selected-token budget. Only then should prefetch, eviction, archived-head conversion, or cache-aware scheduling be optimized.

9 Falsification and Limitations

Deep SGLang integration is unnecessary if RadixAttention plus the tuned black-box path matches native PRA on quality-adjusted memory, TTFT, and throughput. The current baseline is already strong, so a metadata-only object registry cannot satisfy the claim.

The measurements use one small model, a local compatibility patch, one Apple host, one synthetic answer, and five repeats. The model precision differs from the 4-bit runs in the shared figure. HiCache, concurrency, tail latency, resource-affinity routing, archived-head transition, and native detail bytes are unmeasured.

10 Related Work

SGLang combines a structured runtime with RadixAttention for shared prefix reuse [1]. Paged serving systems manage physical KV, while hierarchical caches extend residency across tiers. Retrieval systems select text or documents before generation. PRA differs by assigning stable identity to native non-prefix memory and making its active subset query dependent. The central integration constraint is to preserve both prefix and resource identities rather than collapse one into the other.

11 Conclusion

On the pinned SGLang-MLX environment, selected context and RadixAttention are clearly complementary: the selected condition preserves 5/5 recovery, removes 76.9% of visible prompt tokens, and reuses 364 of 365 tokens on warm requests. That is a useful black-box result. It is not evidence that HiCache can host PRA objects or that SGLang attention can consume selected non-prefix K/V. Paper 6.1 therefore closes the radix-coexistence gate and leaves the engine-native thesis open for a focused hierarchical-object experiment.

References

[1] L. Zheng et al. SGLang: Efficient Execution of Structured Language Model Programs. 2024.